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Enantioselective synthesis of isoxazolines bearing allenes by palladium-catalyzed carboetherification of β,γ-unsaturated ketoximes with propargylic acetates†

Feifei Jin,[a] Cangzhu Hu,[a] Shuaijie Wu,[a] Guodong Zhang,[a] Ying Han,[a] Chao-Guo Yan,[a] Wenbo Li,*[b] Chaoshen Zhang*[c] and Lei Wang*[a]

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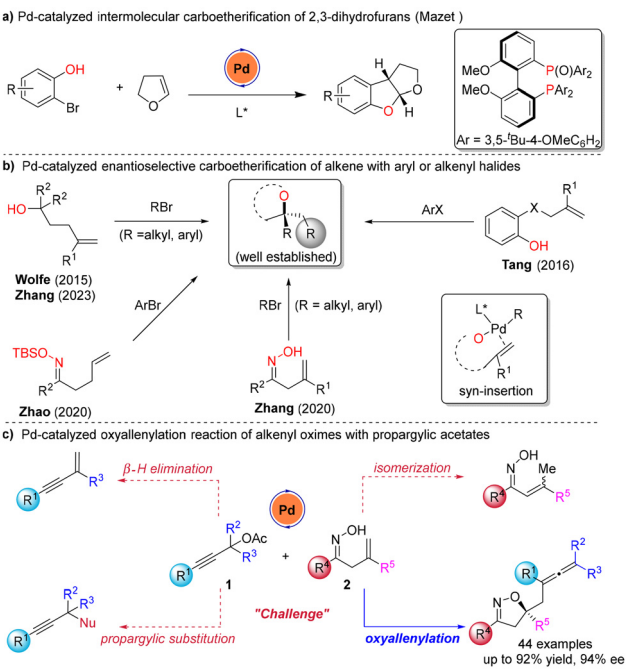
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A highly efficient Pd-catalyzed enantioselective carboetherification reaction of β,γ-unsaturated ketoximes with propargylic acetates is demonstrated. This method provides various chiral isoxazolines bearing multi-substituted allene groups in good yields with excellent enantioselectivities. The sterically electron-rich BINOL-derived phosphoramidite ligand exhibits highly efficient control over both chemoselectivity and enantioselectivity. The salient features of this reaction include readily available starting materials, broad functional group tolerance, an easy scale-up, versatile transformations and promising photo-physical properties. DFT calculations were carried out to disclose the detailed mechanism and origins of the enantioselectivity.

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Introduction

Over the past decades, transition-metal-catalyzed carboheterofunctionalization reactions of alkenes have become a powerful method for the construction of heterocyclic compounds.[1–3] In this regard, palladium-catalyzed carboetherification of alkenes with aryl/alkenyl halides allows the construction of a new C–C bond and a new C–O bond in a single reaction, which has greatly expanded its scope.[4] Despite significant progress, the development of such an asymmetric reaction still remains a challenge, and only a few strategies have been explored so far. For instance, Mazet and co-workers developed an elegant Pd-catalyzed intermolecular carboetherification of dihydrofurans, which provided access to a variety of chiral-fused tetrahydrofurobenzofurans (Scheme 1a).[5] On the other hand, Pd-catalyzed enantioselective carboetherification of alkenes bearing a nucleophilic moiety with aryl/alkenyl halides was also developed.[6–10] For example, the first palladium-catalyzed enantioselective carboetherification reaction of alkenols to afford chiral tetrahydrofuran was realized by Wolfe in 2015 (Scheme 1b).[6] Later, Tang's group developed enantioselective



Scheme 1 Pd-catalyzed asymmetric carboetherification of alkenes.

[a]*School of Chemistry & Chemical Engineering, Yangzhou University, Yangzhou 225002, China. E-mail: lei.wang@yzu.edu.cn*
[b]*State Key Laboratory of Petroleum Molecular & Process Engineering, Shanghai Key Laboratory of Green Chemistry and Chemical Processes, School of Chemistry and Molecular Engineering, EastChina Normal University, 200062 Shanghai, China. E-mail: wbli@chem.ecnu.edu.cn*
[c]*Department of Chemistry, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong SAR 999077, China. E-mail: zhangcs@ust.hk*
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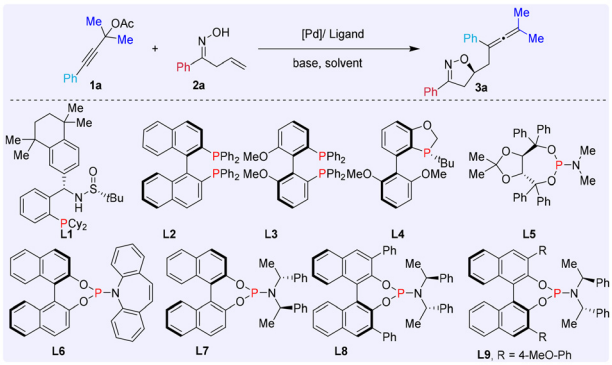
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palladium-catalyzed alkene aryloxyarylation to give a series of chiral 1,4-benzodioxanes, chromans, and 1,4-benzooxazines.[8] Recently, the group of Zhang reported an efficient Pd/Xiang-Phos catalyzed carboetherification of alkenyl oximes to afford isoxazolines with good enantioselectivity.[9] Despite remarkable progress, however, these electrophiles in the carboetherification reaction are limited to aryl/alkenyl halides. With regard to the significance for increasing the diversity of compounds, developing new electrophile variants and widespread applications of specialized structural units would be highly valuable.

Allenes are key structural motifs in natural products and also serve as potential precursors in the synthesis of many bioactive molecules, drugs and target molecules.[11,12] Consequently, the development of novel methodologies to synthesize allenes has attracted enormous attention.[13] Among many strategies, high progress has been achieved in the transition-metal-catalyzed cross-coupling of propargyl electrophiles with various nucleophiles for the preparation of allenes.[14] Remarkably, Zhou and co-workers developed an efficient palladium-catalyzed asymmetric Wacker-type coupling of three-component allenylation reactions.[15] Very recently, our group developed a novel palladium-catalyzed asymmetric alleneamination of β,γ-unsaturated hydrazones with propargylic acetates, which provides facile access to the installation of various multisubstituted allene groups onto chiral dihydropyrazoles.[16] In continuation of our interest in the allenylation reaction, we are curiously interested in whether the reaction might be extended to an asymmetric version of carboetherification of alkenyl oximes with propargylic acetates for the synthesis of allenyl substituted chiral isoxazolines. To achieve this carboetherification reaction, several side reactions must be prevented, and several intricate challenges should be conquered: (1) how to efficiently control the isomerization of β,γ-unsaturated oximes to α,β-unsaturated oximes and prevent propargylic substitution with nucleophilic oximes; (2) minimize β-hydrogen elimination of propargylic acetate atoms to give byproduct enynes; and (3) how to sequentially and efficiently control both chemoselectivity and enantioselectivity in one pot. Herein, we describe an efficient palladium-catalyzed enantioselective carboetherification of alkenyl oximes with propargylic acetates, delivering enantioenriched isoxazolines bearing allenyl groups in good yields with excellent enantioselectivities (Scheme 1c).

At the onset of our study, 2-methyl-4-phenylbut-3-yn-2-yl acetate **1a** with β,γ-unsaturated oximes **2a** were selected as the model substrates to optimize this asymmetric carboetherification reaction. We envisioned that our previously developed palladium-catalyzed asymmetric alleneamination of β,γ-unsaturated hydrazones with propargylic acetates would be capable of controlling the stereochemistry of this asymmetric carboetherification reaction. However, we were disappointed to not observe even traces of the desired product **3a** by using Pd(cod)Cl_2 as the catalyst, **L1** as the ligand, and K_2CO_3 as the base in CPME (cyclopentyl methyl ether) at 50 °C under an argon atmosphere (Table 1, entry 1). Screening of bisphosphine ligands (**L2** and **L3**) and P-chiral ligand **L4** led to the desired product **3a** in poor yields with low enantioselectivities (Table 1, entries 2–4). We then investigated the TADDOL-derived chiral phosphite ligand **L5**, and the desired product **3a** was obtained in 55% yield with a poor ee value (Table 1, entry 5). It was found that BINOL-derived phosphoramidites (**L6** and **L7**) failed to improve yields and enantioselectivities (Table 1, entries 6 and 7). On further screening more sterically hindered ligands (**L8** and **L9**), we found that **L9** was the best ligand, affording **3a** in 82% yield with 89% ee (Table 1, entry 9). With the best ligand (**L9**) in hand, we then focused on exploring various palladium sources. The effect of palladium sources such as $Pd_2(dba)_3$, $Pd(OAc)_2$ and allylpalladium(II) chloride failed to improve the yield and enantioselectivity of **3a** (Table 1, entries 10–12). However, the substrate has relatively good reactivity in THF and toluene, leading to **3a** with a relatively lower ee value (Table 1, entries 13 and 14). Notably, a range of bases such as NaO^{t}Bu, KO^{t}Bu, and Na_2CO_3 did not give good results (for additional details, please see the ESI†). Finally, the enantioselectivity could be further improved to 92% ee by the use of CPME (5.0 mL) at 50 °C under argon for 48 h.

Table 1 Optimization of the reaction conditions[a]

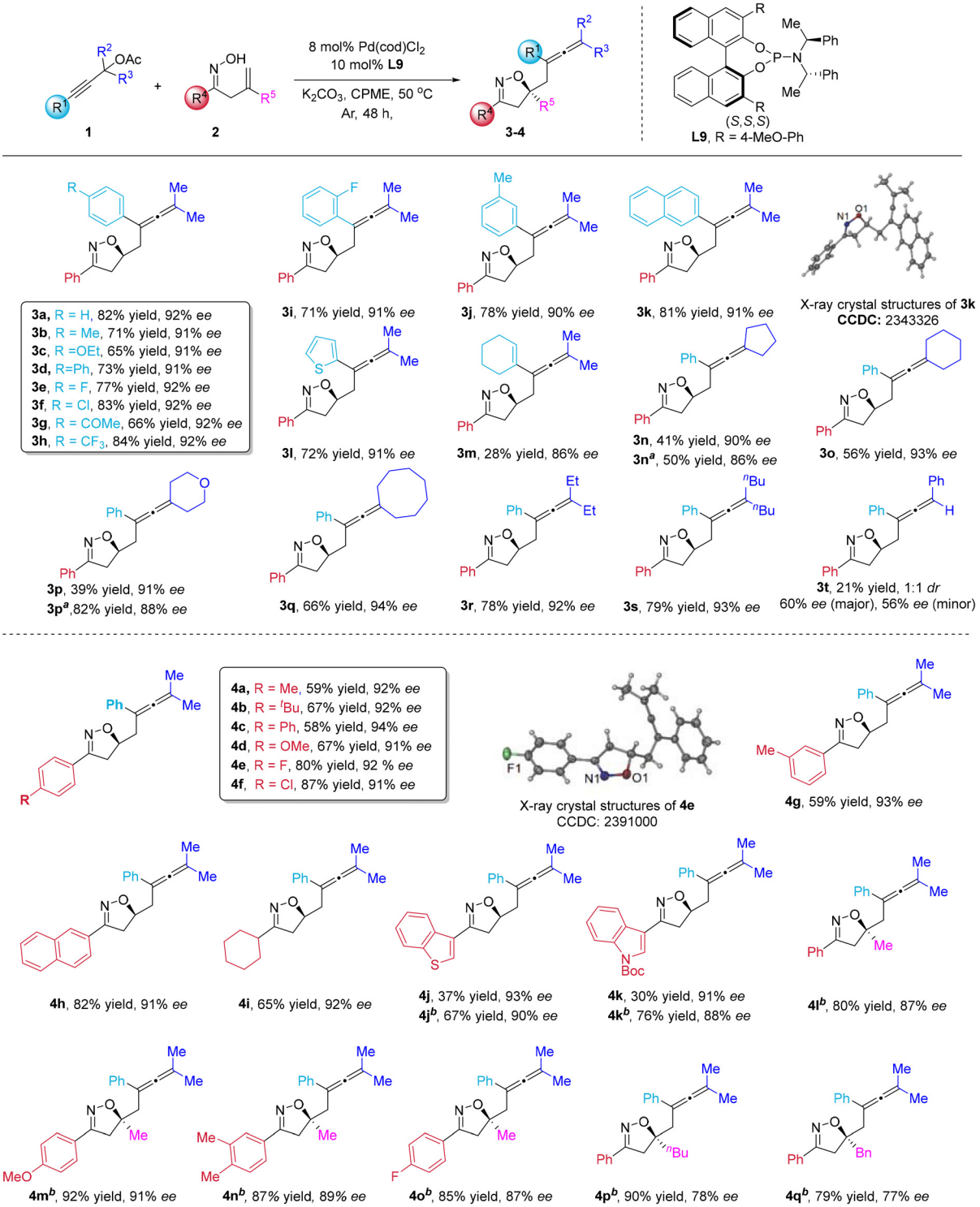


Entry	Pd catalyst	Ligand	Solvent	Yield[b] (%)	ee[c] (%)
1	Pd(cod)Cl_2	**L1**	CPME	NR	NR
2	Pd(cod)Cl_2	**L2**	CPME	5	9
3	Pd(cod)Cl_2	**L3**	CPME	10	<5
4	Pd(cod)Cl_2	**L4**	CPME	28	−19
5	Pd(cod)Cl_2	**L5**	CPME	55	19
6	Pd(cod)Cl_2	**L6**	CPME	10	−14
7	Pd(cod)Cl_2	**L7**	CPME	16	<5
8	Pd(cod)Cl_2	**L8**	CPME	82	85
9	Pd(cod)Cl_2	**L9**	CPME	82	89
10[d]	$Pd_2(dba)_3$	**L9**	CPME	22	90
11	$Pd(OAc)_2$	**L9**	CPME	74	90
12[d]	$[Pd(allyl)Cl]_2$	**L9**	CPME	76	91
13	Pd(cod)Cl_2	**L9**	THF	94	87
14	Pd(cod)Cl_2	**L9**	Toluene	83	71
15[e]	Pd(cod)Cl_2	**L9**	CPME	82	92

[a] Reaction conditions: **1a** (0.2 mmol), **2a** (0.3 mmol, 1.5 equiv.), K_2CO_3 (0.3 mmol, 1.5 equiv.), Pd catalyst (8 mol%) and ligand (10 mol%) in 2.0 mL CPME (cyclopentyl methyl ether) at 50 °C under argon for 24 h. [b] Isolated yield. [c] ee was determined by HPLC analysis. [d] 4 mol% Pd catalyst was used. [e] **1a** (0.3 mmol), **2a** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in 5.0 mL CPME at 50 °C under argon for 48 h.

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Scheme 2 Enantioselective synthesis of isoxazolines. Unless otherwise noted, all reactions were carried out with **1** (0.3 mmol), **2** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol, 1.5 equiv.), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in CPME (5.0 mL) at 50 °C under argon for 48 h. [a] **1** (0.3 mmol), **2** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol, 1.5 equiv.), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in THF (5.0 mL) at 50 °C under argon for 48 h. [b] **1** (0.3 mmol), **2** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol, 1.5 equiv.), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in THF (2.0 mL) at 50 °C under argon for 48 h.

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Results and discussion

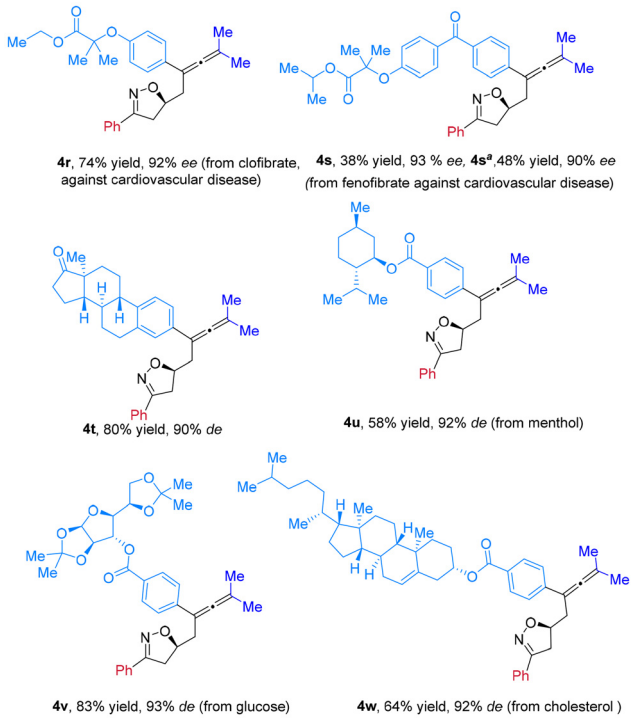
With the optimized conditions in hand, the scope of propargylic acetates **1** was first examined as shown in Scheme 2. For example, a variety of propargylic acetates bearing electron-donating or electron-withdrawing groups at the *para* position of the phenyl ring were all well tolerated, and afforded the desired products **3b–3h** in moderate to excellent yields with 91–92% ees. Notably, substrates with aryl groups bearing *ortho*- and *meta*-substituents were also applicable to give the desired products **3i** and **3j** in 71–78% yields with 90–91% ees.

Moreover, 2-naphthyl propargylic acetates could be employed to deliver the corresponding product **3k** in 81% yield with 91% ee. The absolute configuration of **3k** was confirmed by X-ray crystallography analysis.[17] Heteroarene and 1-ethynylcyclohex-1-ene substituted propargylic acetates also proved to be good substrates in this transformation, affording **3l** and **3m** with comparable enantioselectivities. In addition, the propargylic carbonates containing five-, six- and eight-membered cyclic alkyl scaffolds reacted smoothly to provide **3n–3q** in moderate yields with excellent enantioselectivities (86–94% ees). Substrates with the ethyl or *n*-butyl located on the tertiary carbon atom of the propargylic acetates were also good reaction partners and led to the corresponding products **3r** and **3s** in good results.

Next, we then investigated the scope of unsaturated ketoximes in this asymmetric reaction. A wide range of β,γ-unsaturated ketoximes with phenyl rings containing electron-donating or electron-withdrawing groups at different positions were tolerated well, providing the desired products **4a–4g** in 58–87% yields with 91–94% ees. Moreover, polycyclic aryl, heteroaryl and alkyl-substituted ketoximes could be employed to afford **4h–4k** in moderate yields with good enantioselectivities (88–93% ees). In addition, *gem*-disubstituted alkenes ketoximes were successful in producing **4l–4o** in excellent yields with 87–91% ees. However, replacing the methyl groups with other alkyl groups led to **4p** and **4q** in acceptable yields with relatively lower enantioselectivities of 77–78% ees.

To demonstrate potential applications, late-stage modification of pharmaceuticals and natural products was then conducted (Scheme 3). The clofibrate and fenofibrate derivative products **4r** and **4s** were obtained in moderate yields with high ees. In particular, estrone menthol, carbohydrate and cholesterol derived **4t–4w** were afforded in good yields with high diastereoselectivities.

To further explore the utility of this protocol, we carried out experiments to probe a 5 mmol scale reaction of **1a** under the standard conditions, and the desired product **3a** was obtained in 78% yield (1.18 g) with 92% ee (Scheme 4a). First, allene **3a** allowed the formation of ketone **5** in 79% yield with 91% ee by the use of a $NaIO_4/RuCl_3$ oxidation system. Moreover, the enantioenriched isoxazolines can be transformed into β-hydroxy ketones **6** in 60% yield with 91% ee. In addition, *N*-iodosuccinimide (NIS)-mediated cyclization of **3a** provided the 2-iodoindene framework **7** in 95% yield without loss of enantioselectivity. Subsequently, the product **7** was subjected



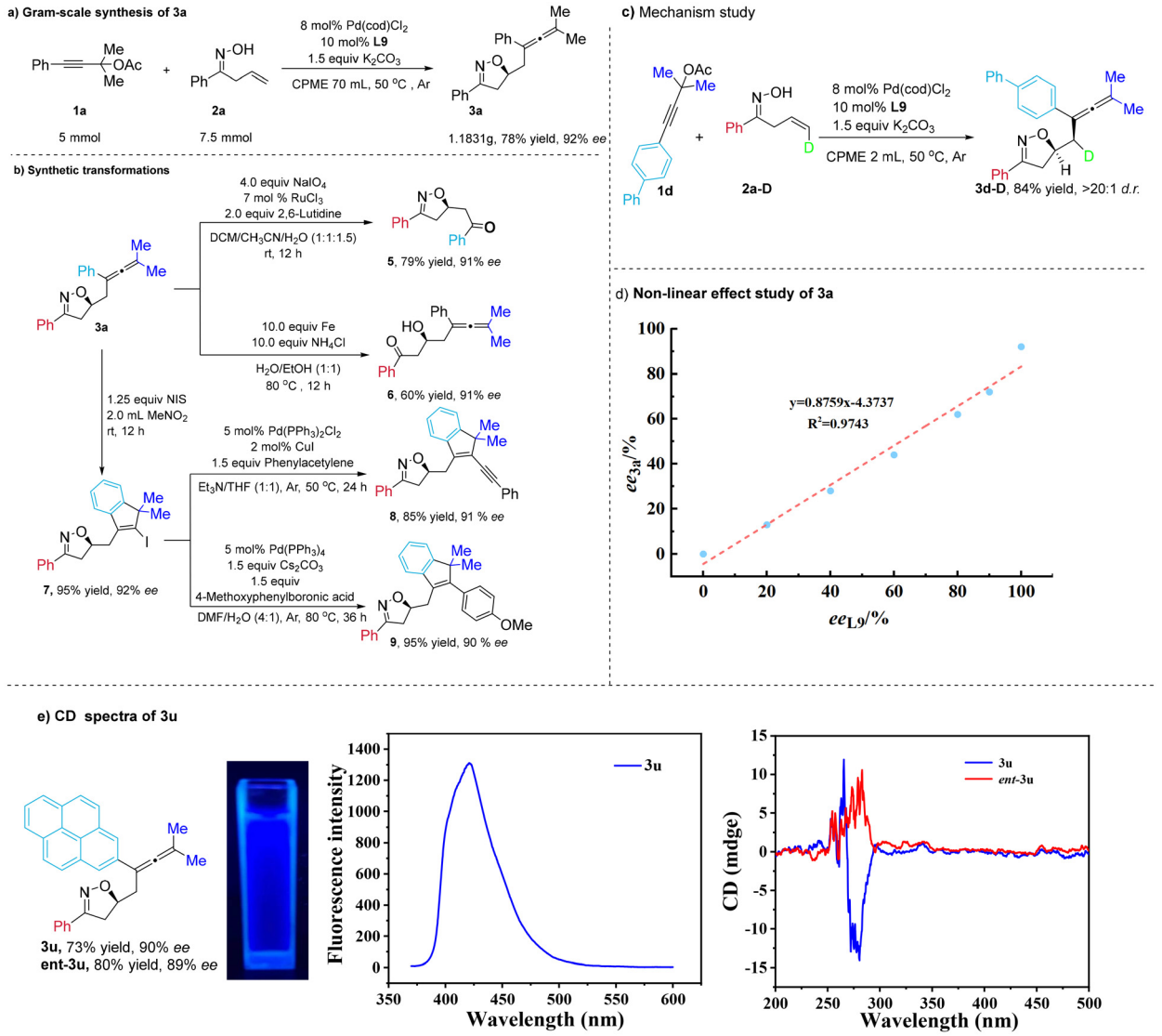
Scheme 3 Late-stage modification of pharmaceuticals and natural products. Unless otherwise noted, all reactions were carried out with **1** (0.3 mmol), **2** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol, 1.5 equiv.), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in CPME (5.0 mL) at 50 °C under argon for 48 h. [a] **1** (0.3 mmol), **2** (0.45 mmol, 1.5 equiv.), K_2CO_3 (0.45 mmol, 1.5 equiv.), Pd(cod)Cl_2 (8 mol%) and **L9** (10 mol%) in THF (5.0 mL) at 50 °C under argon for 48 h.

to palladium-catalyzed Sonogashira or Suzuki–Miyaura reactions to afford products **8** and **9**, respectively, in excellent enantioselectivities (Scheme 4b). More reactions of β,γ-unsaturated ketoxime **2a-D** with propargylic acetates **1d** were conducted, the product **3d-D** was obtained in good yield with >20 : 1 dr, and *syn*-addition pathways were possible in this reaction supported by the J_{H-H} coupling constant (9.4 Hz) (Scheme 4c).[4f] Notably, the linear relationship analysis of the ee values of products **3a** and **L9** with various degrees of enantiomeric excess was conducted, as shown in Scheme 4d, suggesting that monomeric **L9**–Pd might be the actual catalytic species. Moreover, the photo-physical properties of **3u** were further investigated by ultraviolet–visible (UV/vis) absorption, photoluminescence and CD spectroscopy, indicating its potential application in organic optoelectronic materials (Scheme 4e).

To fully understand the catalytic transformation and gain deep insight into the enantiodetermining transition states, density functional theory (DFT) calculations were carried out on the model reaction of propargylic acetates **1a** and β,γ-unsaturated ketoximes **2a**, as shown in Fig. 1 (see the ESI† for details). Initially, the generated pre-cat undergoes substrate–ligand exchange with **1a**, to afford active species **INT1**. Then, Pd(0)L* could undergo oxidative addition with **1a** to

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Scheme 4 (a) Gram-scale synthesis. (b) Synthetic applications. (c) Mechanistic studies. (d) Non-linear effect study of **3a**. (e) Photophysical property investigations: emission spectra of **3u** (10^{-4} (M)) at room temperature (λ_{exc} = 340 nm; slits, 2.5/2.5 nm); CD spectra of **3u** (10^{-4} (M)) were investigated.

form the Pd(II)-complex. A general transition state **TS1″** to break the C–O bond of **1a** was first examined but failed to locate, which prompted us to consider other possibilities. Specifically, an unusual five-membered ring transition state **TS1′** was elaborated, but with an energy barrier of 28.1 kcal mol^{-1} (**TS1′** relative to **INT1**). More importantly, the seven-membered ring **TS1** could significantly reduce the barrier for oxidative addition to 14.7 kcal mol^{-1}, yielding **INT2**. Subsequently, the concerted metalation–deprotonation (CMD) mechanism occurs directly from **INT2** *via* K_2CO_3-mediated **TS2** to deliver **INT3**, and the by-product $KOAc{\cdot}KHCO_3$. It was noteworthy that the double bond in **INT3** could coordinate with the Pd(II) center *via* two possibilities to generate **INT4**, depending on the different coordination directions of the double bond and the Pd-center. The following olefin insertion is the key step to control the enantioselectivity. Therefore, we carefully consider four possible transition states in the process, namely **TS3**-*S*, **TS3**-*S′*, **TS3**-*R* and **TS3**-*R′*, to form the C–O bond. The calculated energy difference (2.2 kcal mol^{-1}) between **TS3**-*S* and **TS3**-*R′* reasonably agrees with the experimental ee value of **3a**. Finally, the resulting **INT5**-*S* undergoes reductive elimination *via* **TS4**-*S* to afford the coupling product **3a** and recover the substrate coordinated-Pd(0) species (**INT1**). Notably, although **TS4**-*S* is not inferior to **TS3**-*S* but not as good as **TS3**-*R′*, we believed that the generated **INT5**-*S* could smoothly pass through reductive elimination instead of returning to **INT4**-*S*, due to the greater thermodynamic driving force backward.

To further understand how the chiral ligand L* induces the enantioselective transition states, we performed detailed structural analyses. As shown in Fig. 1B, in the well-organized **TS3**-*S* and **TS4**-*S*, the unfavorable $H^{\delta+}\cdots H^{\delta+}$ steric repulsion between the chiral ligand L* and the substrates **1a** and **2a** is minimized,

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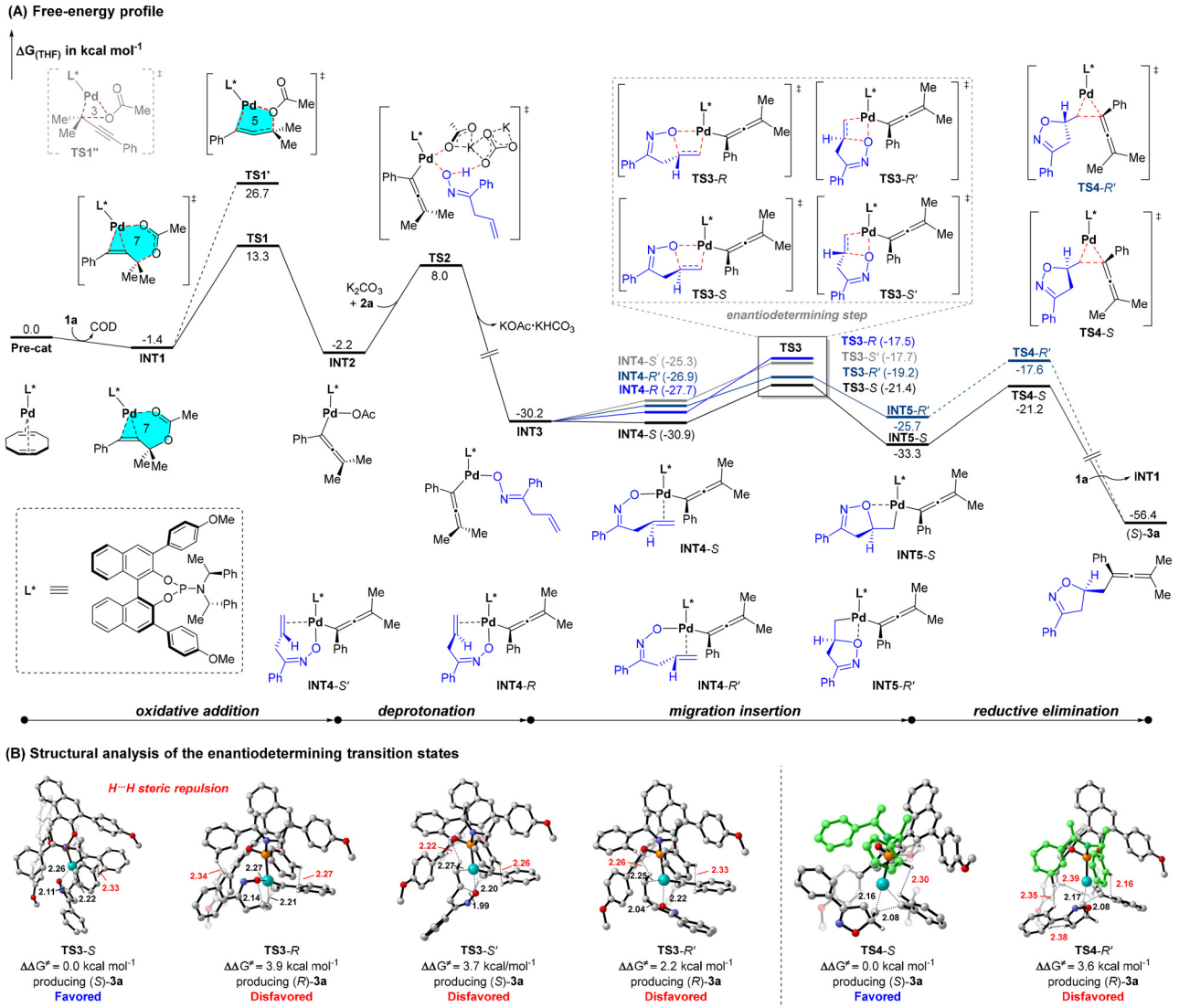


Fig. 1 Free energy profiles (in kcal mol^{-1}) for the enantioselective coupling of **1a** and **2a** *via* the Pd(0)/Pd(II) cycle. The key bond lengths in the enantioselective transition states are given in angstroms.

as reflected in the sum of the H···H van der Waals radius distances of 2.33 Å and 2.30 Å, respectively. In contrast, the corresponding enantioselective isomers are subject to stronger $H^{\delta+}\cdots H^{\delta+}$ steric hindrance, which is the key factor controlling the formation of the major (*S*)-enantiomer in the enantioselective stage.

Conclusion

In summary, we have developed a novel Pd-catalyzed enantioselective carboetherification reaction of β,γ-unsaturated ketoximes with propargylic acetates to afford various chiral isoxazolines bearing multisubstituted allene groups in good yields with excellent enantioselectivities. This reaction offers advantages including a general substrate scope, broad functional group tolerance, good yields and high enantioselectivity. These highly enantioenriched products are precursors of other useful chiral building blocks. The photo-physical properties were further investigated, indicating their potential application in organic optoelectronic materials. DFT calculations further indicated that the stereochemistry was regulated by the chiral ligand-promoted olefin insertion step, wherein the favored transition state is stabilized by less steric hindrance in the chiral ligand and the substrates. Further expansion of the scope of asymmetric Pd-catalyzed reactions is ongoing in our laboratory.

Author contributions

L.W. and W. L. conceived the project. C.Z. and W.L. wrote the paper. F.J. performed the experiments and analyzed the data. C.Z. conducted the computational studies. F.J., C.H., S.W., G.

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Z., Y.H., and C.Y. analyzed the data. All authors discussed the results and commented on the paper.

Data availability

The data supporting this article have been included as part of the ESI.† All detailed procedures, characterization data and NMR spectra are available in the ESI.†

Conflicts of interest

There are no conflicts to declare.

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